FROM FDG TO PSMA: FEW-SHOT LESION SEGMENTATION AND REPRESENTATION TRANSFER IN WHOLE-BODY PET/CT

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ABSTRACT

Whole-body PET/CT lesion segmentation is highly sensitive to tracer chemistry: models trained on FDG rarely transfer to PSMA without target labels, yet PSMA annotations are scarce in many clinical settings. We study this *tracer-shift* problem under a unified aligned protocol that freezes a shared FDG supervised stage and varies only the PSMA adaptation budget ($N \in \{50, 10, 5, 0\}$ labeled cases), plus a high-label PSMA reference (70%) and an FDG held-out test. Across fifteen method variants spanning classical segmentation, self-supervised representation learning, promptable PET models, and competition entries, we find that representation-learning initializations—in particular PET/CT masked autoencoding— retain higher TEST Dice as PSMA labels vanish, while several strong supervised or promptable baselines degrade primarily through false negatives on sparse lesions. We argue that the long-tailed, domain-conditional statistics of clinical findings should shape both pretraining objectives and evaluation (Dice together with voxel-level FP/FN), and release code and a mean-metric board for the aligned protocol.[1]

1 INTRODUCTION

Oncology PET/CT is multimodal and multi-tracer. FDG and PSMA probe overlapping but statistically different lesion populations: prevalence, SUV patterns, organ background, and negative-study rates all shift with the tracer (Chen et al., 2025). In practice this yields a representation-learning problem that is closer to open-world transfer than to i.i.d. fine-tuning: abundant (or at least more available) FDG supervision must support PSMA segmentation when only a handful of PSMA labels exist.

Recent medical foundation / SSL models promise transferable features (He et al., 2022; Tang et al., 2022; Anonymous, 2026), while strong supervised toolkits such as nnU-Net remain hard baselines (Isensee et al., 2021). Concurrent work on normative representation learning further emphasizes that clinical findings are rare, heterogeneous, and imperfectly observed, so objectives and metrics tuned only to frequent lesions can be misleading (Bercea et al., 2025b;a).

This draft asks a concrete empirical question under matched compute and splits: *Given the same FDG supervised stage, which initializations and adaptation recipes remain useful when PSMA labels shrink from 50 to 0 cases?* Our contributions toward a workshop paper are:

1. An **aligned FDG→PSMA protocol** with shared FDG training, PSMA few-/zero-shot ladders, and TEST reporting of Dice / FP / FN.
2. A **multi-method board** covering MIM/nnU-Net, Swin SSL, DpDNet, SegAnyPET, PET/CT MAE, retrieval, and AutoPET V entries.
3. An analysis linking **clinical finding statistics** (tracer shift, label scarcity, FN-dominated failures) to representation objectives and evaluation.

[1] Draft for an ICLR workshop track; numbers are live experimental snapshots and will be frozen for the camera-ready.

Table 1: PSMA/FDG TEST **Dice (%)** under the aligned protocol (higher is better). "–" denotes pending/incomplete runs. Bold marks the best completed entry per column among finished rows.

Method	Init	fs50	fs10	fs5	fs0	fc70	FDG
Proto+Retrieval	retrieval	0.0	0.0	0.0	0.0	0.2	0.0
nnU-Net MIM	MIM	41.6	36.9	32.5	19.1	44.0	61.2
nnU-Net	scratch	32.2	31.6	29.1	15.4	33.0	56.2
MONAI SwinViT	Tang SSL	44.6	29.4	26.1	15.5	56.1	61.0
MONAI SwinViT	scratch	46.1	32.9	29.2	11.5	57.5	61.0
DpDNet dual-enc	dual-enc	45.0	41.4	36.2	17.7	49.2	63.9
DpDNet	scratch	44.0	40.8	**41.2**	16.7	44.0	61.4
SegAnyPET	Lesion	15.6	8.4	8.0	**26.0**	22.0	44.7
SegAnyPET	scratch	8.0	3.7	3.0	22.1	19.2	35.6
PET/CT MAE	MAE SSL	**53.0**	**39.3**	30.6	16.0	**60.5**	**73.2**
PET/CT MAE	scratch	48.1	38.2	31.2	15.6	58.1	66.3
BIRTH	scratch	–	32.5	29.5	–	–	–
BIRTH	MT619	–	–	–	–	–	–
YixinChen	scratch	32.7	29.4	26.6	–	–	–
YixinChen	MT619	–	–	–	–	–	–

2 ALIGNED EXPERIMENTAL PROTOCOL

Data. We use a stratified PET/CT cohort with tracer-aware roles (PSMA/FDG $\times$ small/large/negative lesion burden) and a fixed 70/10/20 train/val/test split seed. PSMA few-shot subsets ($N = 50, 10, 5$) are drawn from the PSMA train pool; PSMA `fs0` evaluates the shared FDG checkpoint zero-shot on PSMA TEST; `fc70%` uses 70% of PSMA train as a high-label reference; FDG TEST probes source-domain retention.

Shared FDG stage. Every non-retrieval method first trains (or fine-tunes) on the same FDG supervised split with matched global batch size and a common early-stopping / epoch budget policy for the board (method-specific defaults are logged). The resulting FDG checkpoint initializes PSMA adaptation unless the method is training-free.

PSMA adaptation. Few-shot fine-tuning uses small labeled PSMA sets with online validation and TEST evaluation aggregated over nine TEST folds (means reported in the main text; per-fold grids omitted). Metrics:

$$\mathrm{Dice} = \frac{2\mathrm{TP}}{2\mathrm{TP} + \mathrm{FP} + \mathrm{FN}}, \quad \mathrm{FP}\% = \frac{\mathrm{FP}}{\mathrm{Neg}}, \quad \mathrm{FN}\% = \frac{\mathrm{FN}}{\mathrm{Pos}},$$

with Dice computed excluding empty-GT cases and FP/FN as voxel micro-averages. This decomposition matters clinically: under sparse PSMA disease, mean Dice can hide near-total miss rates (high FN).

Methods. We compare (i) scratch vs. pretrained siblings when available; (ii) PET/CT MAE SSL (Anonymous, 2026); (iii) Swin transformer SSL (Tang et al., 2022); (iv) nnU-Net and MIM variants (Isensee et al., 2021); (v) DpDNet dual-encoder / scratch (Anonymous, 2025a); (vi) SegAnyPET pretrained / scratch (Anonymous, 2025b); (vii) prototype retrieval (training-free); (viii) AutoPET V competition codebases (BIRTH / YixinChen), scratch and MultiTalent-initialized rows.

3 RESULTS

Representation pretraining helps under scarcity. On PSMA `fs50`, PET/CT MAE with SSL initialization leads completed methods (53.0 Dice), ahead of its scratch twin (48.1) and of dual-encoder / Swin / nnU-Net MIM rows (Table 1, Fig. 1). The gap persists at `fs10`. At `fs5`, DpDNet scratch is competitive on Dice, but MAE SSL still pairs stronger FDG TEST retention (73.2), suggesting the SSL features are not bought by catastrophic forgetting of the source tracer.

Failure modes are statistically asymmetric. Board FP rates are typically $\ll 1\%$, while FN often exceeds 30–70% as N decreases—especially for SegAnyPET adaptations that collapse toward under-segmentation on PSMA few-shot. This matches the clinical prior that missed sparse lesions dominate error under low prevalence, and argues against reporting Dice alone.

FDG → PSMA few-shot · mean TEST metrics (15 methods)
2026-09-04 09:55:35 CST · Dice / FP / FN (%) · no per-fold grids

Method (pretrained)	① FDG (shared)	PSMA fs50	PSMA fs10	PSMA fs5	PSMA fs0	PSMA fc70%	FDG TEST
Proto+Retrieval (ECCV'20) none (retrieval)	N/A	0.05% 4.30% 87.26%	0.05% 4.30% 87.26%	0.05% 4.30% 87.26%	0.05% 4.30% 87.26%	0.23% 17.75% 35.33%	0.05% 5.75% 75.99%
nnUNet MIM (Nat.Methods'21) PET+CT MIM	DONE 169ep	41.57% 0.01% 45.74%	36.94% 0.00% 70.89%	32.45% 0.00% 75.50%	19.12% 0.07% 54.50%	43.96% 0.01% 42.61%	61.24% 0.00% 38.01%
nnUNet (Nat.Methods'21) scratch	DONE 1h06m 169ep	32.19% 0.02% 40.20%	31.60% 0.01% 59.63%	29.14% 0.01% 66.77%	15.44% 0.12% 56.41%	32.96% 0.02% 44.13%	56.23% 0.01% 36.75%
MONAI SwinViT (CVPR'22) Tang SSL	DONE 4h28m 100ep	44.56% 0.05% 45.76%	29.37% 0.03% 76.19%	26.12% 0.03% 78.00%	15.48% 0.33% 57.41%	56.10% 0.04% 36.12%	61.00% 0.02% 37.82%
MONAI SwinViT scratch (CVPR'22) scratch	DONE 4h21m 100ep	46.08% 0.05% 43.02%	32.94% 0.03% 71.17%	29.18% 0.01% 78.25%	11.48% 0.73% 50.85%	57.55% 0.05% 32.48%	61.02% 0.02% 34.31%
DpDNet dual-enc (MICCAI'25) PET+CT dual-enc	DONE 169ep	45.00% 0.02% 36.76%	41.41% 0.01% 55.87%	36.25% 0.01% 64.74%	17.69% 0.12% 47.97%	49.19% 0.01% 33.83%	63.90% 0.00% 29.04%
DpDNet (MICCAI'25) scratch	DONE 2h31m 169ep	43.96% 0.01% 38.89%	40.75% 0.00% 74.57%	41.23% 0.00% 73.66%	16.71% 0.12% 50.23%	44.03% 0.01% 39.39%	61.42% 0.00% 39.16%
SegAnyPET (ICCV'25) SegAnyPET-Lesion	DONE 6h54m 100ep	15.64% 0.00% 97.74%	8.40% 0.00% 99.67%	8.02% 0.00% 99.55%	25.98% 0.01% 90.59%	21.97% 0.00% 95.38%	44.68% 0.00% 72.37%
SegAnyPET scratch (ICCV'25) scratch	DONE 7h43m 100ep	7.95% 0.00% 99.24%	3.67% 0.00% 99.93%	3.03% 0.00% 99.78%	22.13% 0.01% 93.34%	19.16% 0.00% 97.28%	35.59% 0.00% 76.62%
PET/CT MAE (arXiv'26) PET/CT MAE SSL	DONE 4h23m 100ep	53.01% 0.06% 33.31%	39.33% 0.04% 57.04%	30.61% 0.03% 75.41%	15.98% 0.60% 31.24%	60.49% 0.06% 26.23%	73.20% 0.01% 20.62%
PET/CT MAE scratch (arXiv'26) scratch	DONE 4h41m 100ep	48.10% 0.05% 41.34%	38.23% 0.03% 63.46%	31.16% 0.03% 71.79%	15.62% 0.50% 45.44%	58.08% 0.05% 32.01%	66.30% 0.02% 26.99%
BIRTH scratch (AutoPET V'26) scratch	DONE 169ep	RUNNING	32.47% 0.01% 69.80%	29.46% 0.00% 75.78%	–	–	–
BIRTH / hemingduo (AutoPET V'26) Dataset619 MultiTalent	PENDING	–	–	–	–	–	–
YixinChen scratch (AutoPET V'26) scratch	DONE 169ep	32.67% 0.02% 42.20%	29.39% 0.00% 73.87%	26.57% 0.00% 78.01%	–	–	–
YixinChen / chenyixin (AutoPET V'26) Dataset619 MultiTalent	PENDING	–	–	–	–	–	–

Figure 1: Mean TEST board for the aligned protocol (Dice / FP / FN). Nine-fold grids, ETA, and queue footers are omitted. Snapshot timestamp is embedded in the figure header; incomplete rows remain marked pending/running.

Zero-shot is not solved. PSMA `fs0` remains low for almost all learned segmentors (∼11–19 Dice), while SegAnyPET's higher fs0 Dice coexists with very high FN on few-shot columns, underscoring that training-free / promptable behavior and few-shot fine-tuning can optimize different operating points.

4 DISCUSSION: CLINICAL STATISTICS ↔ OBJECTIVES

Three statistical facts about findings in this setting should influence representation learning (Bercea et al., 2025b):

1. **Tracer-conditional normals.** "Healthy" and background uptake are domain-specific; FDG-pretrained features must be judged by PSMA transfer, not only FDG TEST.

2. **Long-tailed lesion burden.** Few-shot and empty-GT rates make purely discriminative objectives brittle; generative / masked objectives that densify anatomical–metabolic structure are a better inductive bias when positives are rare.
3. **Annotation as a noisy observation model.** Overlap metrics against coarse masks can punish conservative segmentors; FP/FN stratification and open-world rare-phenotype tests (Bercea et al., 2025a) are necessary complements.

We view the aligned board as a stress test of whether a representation was learned under assumptions compatible with these statistics.

5 Conclusion

We presented a draft aligned benchmark for FDG→PSMA few-shot lesion segmentation. Early evidence favors PET/CT representation pretraining for label-scarce PSMA adaptation, while highlighting FN-dominated failures that Dice alone conceals. Next revisions will freeze incomplete competition rows, add confidence intervals over folds, and tighten citations for concurrent PET foundation models.

Reproducibility statement

Code, split JSONs, and the mean-metric board exporter are available at `https://github.com/yeebowang/fdg2psma-fewshot`. Raw PET/CT volumes and checkpoints are not redistributed here; scripts document the aligned launchers used to produce Fig. 1 and Table 1.

Ethics statement

This work uses research PET/CT imaging under existing data-use agreements for algorithm development. No identifiable patient information is released in the public repository. Clinical deployment would require prospective validation and institution-specific governance.

AI use statement

Large language model tools were used to help edit LaTeX wording and repository documentation. Experimental design, training, evaluation, and numerical results were produced by the authors' pipelines; the authors verified all reported metrics against logged board JSON.

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A Protocol pointers

Launchers and board ingestion live under `run/` and `scripts/iclr2026_aligned_fdg_fs50_board.py` in the public repository. The README figure omits per-fold grids by design; full fold dictionaries remain in internal board JSON for auditing.